

Representation Sharing Limits Discrimination Recovery in Amortized Neural IRT

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Abstract

Amortized neural item response theory (IRT) trains an encoder to predict responses and reads item discrimination and step thresholds from learned representations. The model predicts well but recovers its parameters at different rates: step thresholds and ability quickly, discrimination slowly and with high variance across random seeds. We show that this difference depends on how the representation is arranged. When discrimination is read from a representation shared with the higher-leverage parameters, it is recovered less accurately than when it is read from its own representation, although the responses contain enough information to recover it: a per-item estimator given the true ability recovers discrimination well. Under a fixed training budget, a decoupled representation, or one that lets discrimination depend on the inferred ability, recovers discrimination more accurately than a shared one. The difference shrinks as data and training grow and disappears in the limit. The effect is specific to discrimination, it is visible in the geometry of the shared representation, where discrimination occupies about a tenth of the variance while the step thresholds dominate, and it appears in a non-psychometric encoder as well, so it is a property of amortized encoders rather than of IRT. A nominal response control, whose slope on ability carries information comparable to its intercept, separates two causes the partial credit model ties together. The sharing trade-off and its decoupling remedy recur for that slope, so they follow from sharing a representation rather than from low information, while conditioning the readout on the inferred ability helps only the low-information parameter. For practice, item discrimination read from a shared representation should be reported only after the discrimination readout is decoupled.

1 Introduction

A single neural encoder now serves two purposes in educational measurement. It reads a learner’s response history and predicts the next response, the task on which deep knowledge tracing is built [1, 2], and, in neural IRT, it also exposes interpretable parameters, so the same forward pass returns a learner’s ability and an item’s step thresholds and discrimination [4, 5]. The second purpose is the demanding one. The parameters are meant to be recovered, in the sense that the recovered values rank items as the true values do, and prediction accuracy does not guarantee this.

Recovery is in fact uneven. Trained only to predict, an amortized neural IRT model recovers step thresholds and ability quickly and well, and discrimination slowly, incompletely, and with large variance across random seeds. A model can rank

items by difficulty correctly and rank them by discrimination poorly while predicting responses equally well in both cases. The question is why one of the three parameters, read by the same network from the same responses, is the one left behind.

A rate argument explains part of this. In gradient training the directions with the most leverage on the loss move first [21, 26], and IRT makes leverage explicit. The sensitivity of the prediction to discrimination is $\partial p / \partial \alpha = (\theta - \beta) w$ with $w = p(1 - p)$, which is zero exactly where the response weight w is largest, near $\theta = \beta$; the sensitivity to ability is $\partial p / \partial \theta = \alpha w$. Discrimination is the low-leverage parameter, so it is learned last. This is correct and it is incomplete. It concerns a parameter on its own, it predicts a slower rate rather than a recovery that settles below what the responses allow, and it says nothing about the item embedding that actually carries discrimination, which the discrimination and step-threshold heads both read and which also feeds the encoder.

That representation is the subject of this paper. We show that, at the training budgets used in practice, discrimination is recovered better when it is read from its own representation than when it shares one with the higher-leverage parameters, and that the difference is not caused by a shortage of information. Three results establish this. A per-item estimator given the true ability recovers discrimination from the same responses, so the responses are informative enough. Holding the responses, the true parameters, and their information fixed, and changing only whether the representation is shared, changes how well discrimination is recovered. And the shared representation carries a measurable trace of the difference: discrimination occupies a small fraction of its variance while the step thresholds occupy most of it. Decoupling the discrimination readout, or letting it depend on the inferred ability, removes most of the difference, and the difference shrinks toward zero as data and training grow.

A nominal response control separates two things the partial credit model ties together. The nominal model gives each response category a slope on ability whose information is not low, unlike partial-credit discrimination. That slope still pays the shared-width trade-off and is still rescued by a separate representation, so the sharing effect follows from sharing a representation, not from low information. Conditioning the readout on the inferred ability, by contrast, helps the partial-credit discrimination but degrades the nominal slope, so that remedy is tied to low information.

The same behavior occurs without any IRT content. A non-psychometric sequence model, with an encoder and a deliberately low-leverage readout, reproduces it, and a variant with no encoder does not. Neural IRT is the case where leverage is available in closed form, so the parameter that will be recov-

ered worst is known before training.

Contributions.

1. At fixed responses and information, whether the representation is shared changes how well discrimination is recovered, and a per-item estimator given the true ability recovers what the shared model does not.
2. The shared representation carries little of the discrimination signal, the step thresholds dominate it, and the higher-leverage parameters do not benefit from a separate representation.
3. Decoupling and ability-conditioning each improve discrimination recovery while the high-leverage control does not move, and the improvement survives a regularization control that removes a separate ability-overfitting effect.
4. A nominal response control separates sharing from information: the sharing trade-off and its decoupling remedy recur for a slope whose information is not low, while ability-conditioning helps only the low-information parameter and degrades the slope whose information is not low.
5. A non-IRT encoder reproduces the effect, and item discrimination read from a shared representation should be reported only after decoupling the readout.

2 Background and Related Work

Prediction and recovery in neural IRT. Deep knowledge tracing models a response sequence with a recurrent or attention encoder and is evaluated by next-response prediction [1, 2, 3]. Neural IRT keeps the encoder and adds an IRT-structured decoder, so the state gives ability while item heads give step thresholds and discrimination [4], and variational and amortized variants estimate item parameters jointly [5, 6]. The closest architecture precedent trains an encoder-decoder neural IRT model and reads item parameters from learned representations [7],¹ but it does not separate recovery by parameter or examine the representation. A variational neural IRT model reports the same weaker recovery of discrimination and attributes it to the variational objective rather than to the loss curvature [8].² These models are evaluated by prediction; where they report recovery, they report it for ability, hold discrimination fixed, or read all parameters from one shared representation without examining how that sharing affects each parameter. That question is the subject here.

¹[VERIFY] Converse, Pu, and Oliveira, AIED 2021 (LNCS 12749), cited here as the closest encoder-decoder neural-IRT architecture precedent. Confirm the architecture and that it does not separate recovery by parameter or study the shared representation.

²[VERIFY] Ma et al. 2024 (arXiv:2310.12010), cited here as reporting weaker discrimination recovery under a variational objective and blaming the variational gap. Confirm the exact claim before submission; we find the deficit under a plain prediction loss with no variational gap.

Polytomous IRT and Fisher information. The generalized partial credit model [9, 11] describes an item by a discrimination and a vector of step thresholds, and gives the information available about each parameter in closed form. Discrimination’s information carries a squared lever arm that is small exactly where the responses are most informative, so discrimination is the parameter with the least information in this model. The nominal response model [10] instead gives each category its own slope on ability and intercept, and there the slope on ability has information comparable to the intercept, which we use as a control. Because the IRT scale is identified only up to an affine transformation, recovery is measured by rank rather than by raw magnitude [11].

The amortization gap. Amortized inference replaces a per-instance optimization with a single network, at a measurable cost in quality [12, 13]. Our result is the version of that cost that falls on parameter recovery and is uneven across parameters. The nearest psychometric models amortize item factor analysis [14] or joint estimation [15]; they read all parameters from one shared representation and report recovery without separating the parameters, which is what we do here.

Shared representations across multiple readouts. A representation trained for several readouts of unequal leverage is used unevenly, a known property of multi-task learning [16, 17]. The common responses rebalance or project gradients so one task does not dominate [18, 19], and compatibility analysis predicts that mismatched co-trained outputs degrade a shared representation [20]. We use a structural response instead, separating the representation, and we find that the readout gradients are orthogonal, so the cost is how the shared representation is used rather than direct gradient conflict. We are not aware of this question studied for the recovery of interpretable measurement parameters.

Learning dynamics. That recovery follows leverage is consistent with the dynamics of gradient learning, in which deep linear networks learn modes in order of their singular value [21, 22], with spectral and kernel analogs [26, 27], and the few large eigenvalues of the Hessian correspond to the high-leverage directions [28]. Three lenses give the same ordering and we borrow the ordering rather than a law from any one of them. The natural gradient rescales updates by the inverse Fisher information, so under ordinary gradient descent the low-information directions move more slowly [23]. Spectral analyses of learning recover modes in order of their singular value or frequency [21, 24]. And many fitted models are sloppy, with a few stiff directions and many soft ones set by the same curvature [25]. These hold in linear or near-optimum regimes; our setting is feature learning rather than the lazy regime, since the encoder’s representation changes during training [29], so we take the leverage ordering as motivation and establish the effect empirically.

Inductive bias and probing. A parameter that is recoverable from the responses but not recovered by the trained model is consistent with the result that latent factors are not recovered without an inductive bias [30], and that a structural constraint promotes separation [31]; our per-item estimator is a probe on the encoder output [32].

3 Method

Model. An encoder E_ϕ reads an interaction sequence and produces a per-learner ability state θ_t . Each item j has a learned embedding e_j that feeds the encoder and is read by heads for discrimination α_j and step thresholds β_j . A GPCM decoder maps $(\theta_t, \alpha_j, \beta_j)$ to a distribution over ordered categories, and training minimizes the response-prediction loss; no parameter is a supervised target.

Leverage. For the binary special case ($K = 2$), with $z = \alpha(\theta - \beta)$, $p = \sigma(z)$, and residual $r = p - y$, the per-parameter gradients are $\partial_\theta L = r\alpha$, $\partial_\beta L = -r\alpha$, $\partial_\alpha L = r(\theta - \beta)$, and the single-response Fisher informations are $I(\theta) = I(\beta) = \alpha^2 w$ and $I(\alpha) = (\theta - \beta)^2 w$ with $w = p(1 - p)$. The GPCM forms (Appendix A) replace the residual by the per-category residual and the lever arm $(\theta - \beta)$ by a category-weighted version, and reduce to these at $K = 2$; in both, the information about discrimination carries the squared lever arm $(\theta - \beta)^2$, which is smallest where the response weight w is largest. So on aggregate discrimination has less information than ability or the step thresholds, and it is identified before training as the parameter recovered least accurately.

A control whose slope is not low information. To separate the low information of discrimination from its being a slope read from a shared representation, we add the nominal response model [10] as a control. It gives each category its own slope on ability $a_{i,k}$ and intercept $c_{i,k}$, with $P(Y_i = k | \theta) = \text{softmax}_k(a_{i,k}\theta + c_{i,k})$ and per-category residual $r_k = p_k - y_k$; through the category logit the gradients are $\partial_{a_k} L = r_k \theta$ and $\partial_{c_k} L = r_k$ (Appendix B). The slope a_k is the analogue of GPCM discrimination, a slope on ability, but its information is not low: the θ^2 multiplier averages to about one for $\theta \sim \mathcal{N}(0, 1)$, so $I(a_k)/I(c_k) \approx 0.90$ (3 seeds), not the several-fold gap discrimination has in GPCM. The nominal model holds the representation arrangement fixed while removing the low information, so we can ask whether the slope lags for being a slope in a shared representation or for its low information.

Arrangements and controls. Two arrangements are varied on this one model: *shared* versus *decoupled*, whether discrimination reads e_j or a separate embedding, and *static* versus *dynamic*, whether discrimination may read the inferred state. The controls are also changes to this model: a per-item maximum-likelihood *oracle* given the true ability and no shared embedding, the step thresholds as a high-leverage control, a *no-encoder* variant, and *LayerNorm* on the encoder. The nominal response control uses the same shared, decoupled, static, and dynamic arrangements, and adds two one-sided decouplings that give only the slope, or only the intercept, its own wide embedding. Recovery is the sign-aligned rank correlation between recovered and true values, used throughout for the rank claims; magnitude is reported separately by linking recovered discrimination to the true scale and reporting the regression slope and RMSE (Section 5).

Non-psychometric model. To separate the arrangement from the IRT task, we use a sequence model with no IRT content: a recurrent encoder infers a per-instance state, and a high-leverage location and a low-leverage slope are read from

a shared item embedding that also feeds the encoder, with leverage set by a knob. The same shared-versus-decoupled change and the same oracle apply.

What the theory settles, and its limits. Three statements, each scoped. First, near an identifiable, correctly specified optimum, gradient flow linearizes to $\dot{\delta} = -F\delta$ with F the Fisher information, so each eigenmode of F decays at a rate equal to its eigenvalue, and discrimination is the small-eigenvalue mode; this is the rate argument, and it assumes proximity to the optimum and a near-zero-residual fit. Second, when the item parameters (α, β) form a free table that can reach the optimal prediction, every gradient is linear in the residual and the gradients vanish together, so the optimum recovers the true ranks up to the affine gauge, for any readout arrangement, which we confirm numerically; this applies to the item table and not to the encoder’s θ . Together with the oracle, which recovers discrimination from the same finite responses, this places the finite-budget difference in the training trajectory rather than in the information the responses carry. Third, any C^1 strictly monotone positive link applied to discrimination preserves the rank of α , so the recovered ranks do not depend on the link; near the optimum it rescales the discrimination learning rate by a constant that a matched learning rate absorbs, so the choice of smooth positive link is not the cause.

We do not claim a conditioning-number law. When we break the collinearity between the Fisher conditioning number and the category count, by varying the conditioning number about $45\times$ at fixed category count, the difference between shared and decoupled is flat, so it does not track the conditioning number. The difference grows with the category count through ordinal information, not through conditioning.

Protocol. Responses are generated from a known GPCM ($N = 800$, $Q = 60$, $T = 60$, $\theta \sim \mathcal{N}(0, 1)$, $\alpha \sim \text{LogNormal}(0, 0.3)$). The encoder is an LSTM, and the results replicate on Transformer and DKVMN encoders. Each reported comparison uses at least ten random seeds with bootstrap confidence intervals and paired across-seed tests, except the nominal response control (Section 4.5), which uses eight seeds at $K = 4$.

4 Experiments

4.1 Recovery follows Fisher information

The recovery curves match the rate argument (Figure 1): ability and the step thresholds reach high rank early, and discrimination lags by hundreds of epochs. The rest of this section asks whether the discrimination lag is set by the information in the responses or by the representation, and what changes it.

4.2 The information is sufficient, and the arrangement changes recovery

The responses are informative enough. If the difference were a shortage of information, no estimator should recover discrimination from these responses. A per-item estimator

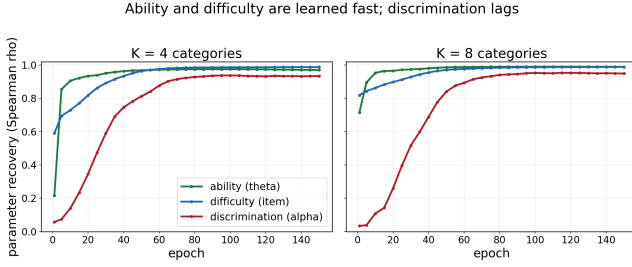


Figure 1: Recovery follows leverage: ability and the step thresholds rise quickly, discrimination slowly.

leverage	shared	decoupled	difference	oracle
high	0.92	0.93	+0.01	0.80
mid	0.85	0.89	+0.04	0.92
low	0.55	0.70	+0.15	0.83

Table 1: Non-psychometric model: recovery of the low-leverage readout ($|\rho|$, 5 seeds). The oracle recovers what the shared model does not, and decoupling improves recovery as leverage falls.

given the true ability does recover it, at 0.83 where the shared model reaches 0.55 at the low-leverage setting of the non-psychometric model (Table 1). The responses carry the discrimination signal, and the shared model does not use all of it.

The arrangement changes recovery at fixed data. If the difference were representational, changing only the arrangement, with the responses and parameters fixed, should change recovery. It does. Widening the shared item embedding in neural IRT traces a trade-off in which ability recovery falls from 0.97 to 0.88 as discrimination rises from 0.66 to 0.91, and no width recovers both; a separate embedding recovers ability near 0.97 and discrimination between 0.90 and 0.94 at the same total capacity, and holds this as capacity grows (Figure 2). The difference is in how the representation is used, not in how much of it there is.

4.3 Geometry of the shared embedding

The shared representation carries little discrimination. If capacity is spent on the higher-leverage parameters, discrimination should occupy little of the shared embedding, and the high-leverage parameters should not benefit from a separate embedding. A linear probe of the shared item embedding recovers the step thresholds at $R^2 = 0.96$ and discrimination at $R^2 = 0.64$, and discrimination accounts for about 11% of the embedding’s variance (Table 2); the step thresholds gain nothing from a separate embedding.

Mechanism and controls. The shared item embedding comes to favor the higher-leverage parameters during training. Under sharing, discrimination recovery rises to 0.92 near epoch 100 and falls to 0.85 by epoch 500, in all ten seeds and as an internal maximum in each, while the separate embedding

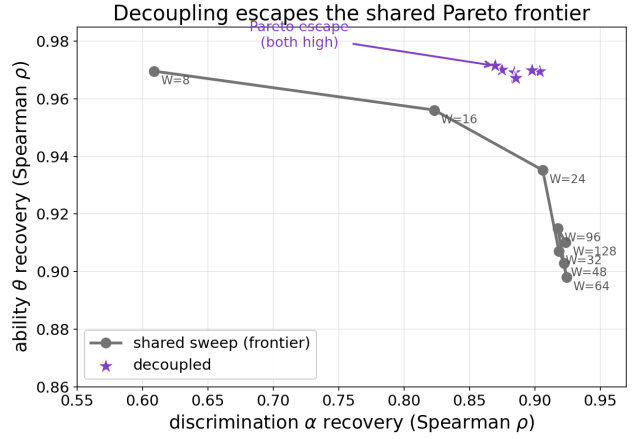


Figure 2: A shared item embedding trades ability against discrimination as its width changes; a separate embedding recovers both at the same total capacity.

	probe R^2, α	probe R^2, β	α var. frac.
shared	0.64	0.96	0.115
decoupled	—	0.86	0.013

Table 2: Geometry of the shared item embedding (≥ 10 seeds). The step thresholds dominate; discrimination accounts for about 11% of the variance. The decoupled α probe is undefined because α no longer lies in the shared embedding.

risks and holds (Figure 3). The gradient on the shared embedding from the ability pathway grows about $30\times$ over training (95% CI [20, 43]) while the discrimination pathway stays flat, and the two are orthogonal (cosine near zero), so the cost is not direct gradient conflict but the use of the shared embedding for the higher-leverage parameters, as the geometry above shows. Regularization separates this from ordinary overfitting: Layer-Norm reduces a distinct ability decay (from -0.18 to -0.06) but leaves the discrimination rise-and-fall unchanged (-0.075 to -0.074). The difference is finite-budget: at a fixed budget it does not shrink as data grow (Figure 4), and it disappears only as data and training grow together. By the linking slope, shared discrimination is also attenuated in magnitude relative to a decoupled or per-item estimate.

4.4 Interventions and the category count

Two interventions improve recovery, and the control is unaffected. Giving discrimination its own embedding is the comparison above. Letting discrimination depend on the inferred ability improves its recovery for $K \geq 4$, by an amount that grows with the number of categories, significant across twelve seeds (paired Wilcoxon $p \leq 0.007$, confidence intervals excluding zero), with no effect at $K = 2$ (Table 3). The step thresholds, which share ability’s leverage ($I(\beta) = I(\theta) = \alpha^2 w$), are unaffected by either change, so the improvement is specific to the low-leverage parameter and is not a general effect of added

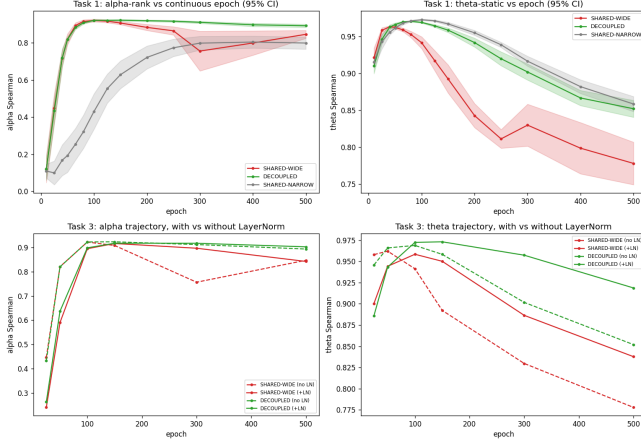


Figure 3: Discrimination recovery over training (10 seeds, 95% CI). The shared embedding rises then falls; the separate embedding holds. LayerNorm removes a separate ability decay but not the discrimination rise-and-fall.

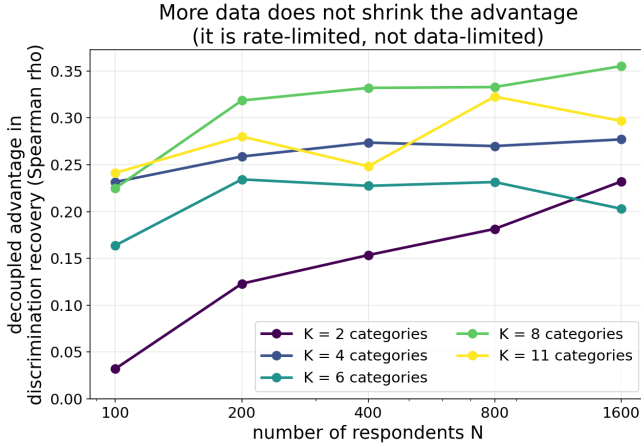


Figure 4: At a fixed training budget the shared-decoupled difference does not shrink as data grow.

flexibility.

4.5 The nominal response dissociation

The partial credit model ties two properties together, a slope on ability and a low information, so the discrimination lag could come from either. The nominal response model separates them: its slope a_k is a slope on ability whose information is not low ($I(a_k)/I(c_k) \approx 0.90$). Trained the same way, on $N = 800$, $Q = 60$, $K = 4$, at 150 epochs over eight seeds, it tells the two apart (Table 4).

The slope is not the slow one. At a shared width of 8 the slope and the intercept recover alike, the slope a_k at 0.85 and the intercept c_k at 0.89, and a separate wide key raises both to 0.980 and 0.982. There is no ordering gap of the kind discrimination shows in GPCM. The lag came from low information, not from being a slope.

The shared trade-off is representation. Widening the

K	Δ_α (95% CI)	Wilcoxon p	Δ_β
2	-0.003 [-0.04, +0.03]	0.68	+0.000
4	+0.032 [+0.01, +0.06]	0.005	+0.001
8	+0.059 [+0.03, +0.09]	0.001	+0.000
11	+0.186 [+0.12, +0.25]	0.0005	+0.000

Table 3: Ability-conditioned versus static discrimination (12 seeds, paired). The improvement grows with K ; the step-threshold control is unaffected.

arrangement	a_k (slope)	c_k (intercept)	ability
shared, width 8	0.85	0.89	0.80
shared, width 32	0.94	0.90	0.62
decoupled (wide key)	0.98	0.98	0.85
+ dynamic slope	0.74	0.97	0.86
decouple c_k only	0.96	0.98	0.87
decouple a_k only	0.87	0.39	0.81
separate keys, both	0.98	0.97	0.84

Table 4: Nominal response model recovery ($|\rho|$, 8 seeds; $N=800$, $Q=60$, $K=4$, 150 epochs). Widening the shared embedding raises the slope a_k but lowers ability; a separate wide key recovers slope, intercept, and ability at once. Conditioning the slope on the inferred state lowers its recovery, although its information is not low. Decoupling the intercept c_k alone leaves the slope high on the narrow value, while decoupling the slope alone breaks the intercept, so the intercept carries the wide-key capacity while the slope shares it.

shared embedding raises the slope by +0.093 from width 8 to width 32 but lowers ability by -0.177, the same trade-off the partial credit model shows, and it appears even though the nominal slope’s information is not low, so it tracks the slope sharing a representation with ability rather than the information being low. A separate wide key clears it, recovering the slope at 0.980, the intercept at 0.982, and ability at 0.853, all above the best shared width at once. Within the nominal model only the intercept needs the wide key: decoupling the intercept c_k alone leaves the slope at 0.964 on the narrow value, decoupling the slope a_k alone drops the intercept to 0.392, and separate keys for both add nothing (0.978 and 0.973). The intercept carries the wide-key capacity while the slope shares it.

Ability-conditioning helps only the low-information parameter. Conditioning the readout on the inferred state rescued GPCM discrimination, which has low information, but it lowers the nominal slope, whose information is not low, from 0.980 to 0.735. The conditioned slope is stable across learner halves, split-half about 0.999, while its recovery falls to 0.735, so where the information is not low the state adds noise rather than signal. The remedy is tied to low information, not to the sharing.

Two mechanisms are now separated. The shared trade-off and the decoupling escape recur for the nominal slope, so they are properties of sharing a representation and hold whether or not the information is low. Conditioning on the inferred ability, by contrast, helps only where the information is low, so

it splits the two models. Discrimination in GPCM has both at once, low information and a slope sharing the step thresholds’ representation; the nominal slope has only the second.

4.6 Recovery versus reliability on real data

On a real corpus there is no ground truth for the item parameters, so we use split-half reliability, the agreement of a recovered parameter across two random halves of the learners, as a proxy for recovery. On EdNet and KDD Cup 2010, discrimination is recovered far less reliably than the step thresholds, EdNet discrimination about 0.79 against step thresholds about 0.94, KDD about 0.63 against about 0.85, and it is under-allocated in the shared embedding, occupying about 1.5% of its variance against the step thresholds’ about 2.5%. The dynamic head is again the lever that does not cost prediction: it is at least as reliable as the static head in every cell and it rescues the thin static decoupled channel, which otherwise collapses (KDD discrimination reliability about 0.39), by about 0.15 on EdNet and about 0.25 on KDD. Decoupling alone can hurt on real data; the dynamic head does not.

[PLACEHOLDER: classical cross-calibration] We will fit a classical GPCM calibration by marginal maximum likelihood (an EM estimator) on the items with the highest response counts and use it as an external reference for discrimination, then test whether the neural-recovered discrimination agrees with it in rank (Spearman) and whether the decoupled or dynamic configurations agree better than the shared one. The classical calibration assumes a static ability, so the comparison is restricted to items and windows where that assumption is defensible, and disagreement that tracks ability change is reported separately. This turns the real-data claim from reliability alone into agreement with an external calibration.

On EdNet option tracing, where each interaction records the chosen option and there is again no ground truth, the decoupling escape depends on coverage. At about twelve observations per item the escape reverses, with shared ability reliability about 0.68 above decoupled about 0.29. At knowledge-component density, about 845 observations per unit, it revives, with decoupled about 0.66 against shared about 0.70. The dynamic head does not rescue at that density, at about 0.49 below the static about 0.66. The escape needs coverage; that the dynamic head still does not help here is the portable finding.

4.7 Downstream utility

[PLACEHOLDER: downstream utility] We will run a computerized adaptive testing simulation that selects each next item by maximum Fisher information at the running ability estimate, using the recovered discrimination and difficulty from each configuration (shared, decoupled, static, dynamic) and the per-item oracle as the ceiling, and scores each by the ability error against the true value at a fixed test length. A companion item-screening test flags the truly low-discrimination items from the recovered estimates. The simulation will show whether the more accurately recovered discrimination of the decoupled and dynamic configurations yields lower ability error or shorter

tests, and how far each sits below the oracle ceiling.

4.8 The effect is not specific to IRT

The non-psychometric model reproduces all of this: decoupling improves the low-leverage readout, the oracle recovers what the shared model does not, the high-leverage readout does not benefit, and decoupling reduces the across-seed variance (Table 1). A variant with no encoder shows no decoupling benefit, which places the effect in the encoder’s shared representation rather than in shared readouts in general. Neural IRT is the case where leverage is available in closed form, so the parameter that will be recovered worst is known before training.

5 Boundaries of the Claim

The evidence for the dynamics is synthetic. On real data we report reliability, the agreement of a recovered parameter across learner halves, not recovery against externally calibrated values; a classical cross-calibration is specified but not yet run (Section 4.6), so the real-data claim is reliability plus a pending calibration, not ground-truth recovery. The result is a finite-budget difference, not a quantitative law: the dependence on data versus training differs between the non-psychometric model and neural IRT, and there is no conditioning-number rate law. The theory is known in parts, the rate argument near the optimum, the free-parameter-table invariant, and the link invariance, each established separately, and we present it as a translation of these into the recovery setting rather than a new dynamics law. We tested and rejected several stronger statements and keep them on record: the link is not the cause, there is no population-limit dynamics law, the conditioning number is not the cause, the recovered-discrimination error does not correlate with ability or step thresholds, and the effect requires the encoder rather than appearing for shared readouts in general.

6 Discussion

The result is the per-parameter form of leverage-ordered learning [21] and the parameter-recovery form of the amortization cost [13], and it leaves classical IRT estimation in place: the model is identified and the estimator is consistent, and what we add is a property of the amortized estimator at finite budgets. It is not gradient conflict, since the readout gradients are orthogonal; where multi-task methods rebalance gradients [18, 19], we separate the representation. The separation is reminiscent of weight normalization, which separates a magnitude from a direction [33], and here it has a known target, since IRT identifies the low-leverage parameter in advance.

The nominal response control shows these are two separate mechanisms. Sharing a representation sets the width trade-off and the decoupling remedy, and applies to any slope read from the shared embedding whether or not its information is low. The information sets the recovery rate and whether con-

ditioning the readout on the inferred ability helps, which is why the dynamic head rescues the low-information discrimination of GPCM but degrades the higher-information slope of the nominal model. Discrimination in GPCM carries both at once, a low information and a slope sharing the step thresholds' representation; the nominal slope carries only the second.

For practice the consequence is direct. Discrimination recovered from a shared neural IRT model is recovered less accurately, less stably across seeds, and attenuated in magnitude; decoupling the discrimination readout, or conditioning it on the inferred ability, improves the recovered ranking at no cost to prediction. Because IRT identifies the low-leverage parameter in advance, the change can be made by design rather than found by trial.

7 Conclusion

Trained to predict, an amortized neural IRT model recovers its parameters at different rates, and discrimination is recovered least accurately, not because the responses are uninformative but because, at finite training budgets, it is read from a representation shared with the higher-leverage parameters. The difference is specific to discrimination, it is visible in the geometry of the shared representation, and it is removed by giving discrimination its own representation or the inferred ability. A nominal response control separates the two mechanisms: sharing a representation sets the trade-off and its decoupling remedy for any slope, while conditioning on the inferred ability helps only the low-information parameter. The effect appears without IRT content, so it is a property of amortized encoders, and neural IRT is the case where the affected parameter is known in advance. Item discrimination read from a shared representation should be reported only after the readout is decoupled.

A GPCM gradients and information

For K ordered categories, let p_k be the GPCM category probabilities, $R_k = p_k - y_k$ the per-category residual, and $B_k = \sum_{c \leq k} \beta_c$. The per-parameter gradients are $\partial_\theta L = \alpha \sum_k R_k k$, $\partial_\alpha L = \sum_k R_k (k\theta - B_k)$, and $\partial_{\beta_c} L = -\alpha \sum_{k \geq c} R_k$, and the single-response Fisher informations are $I(\theta) = \alpha^2 \text{Var}_p(k)$, $I(\alpha) = \text{Var}_p(k\theta - B_k)$, and $I(\beta_c) = \alpha^2 P_{\geq c}(1 - P_{\geq c})$. These reduce to the binary forms in Section 3 at $K = 2$, and in all cases the information about discrimination carries the squared category-weighted lever arm.

B Nominal response gradients and information

The nominal response model gives item i category slopes $a_{i,k}$ and intercepts $c_{i,k}$ with $P(Y_i = k | \theta) = \text{softmax}_k(a_{i,k}\theta + c_{i,k})$ and per-category residual $r_k = p_k - y_k$. Through the category logit $z_k = a_k\theta + c_k$, with $\partial_{a_k} z_k = \theta$ and $\partial_{c_k} z_k = 1$, the

gradients are $\partial_{a_k} L = r_k \theta$ and $\partial_{c_k} L = r_k$. The single-response information about a slope carries the factor θ^2 relative to the matched intercept, so under $\theta \sim \mathcal{N}(0, 1)$ the two informations are comparable in expectation, $\mathbb{E}[\theta^2] = 1$; empirically $I(a_k)/I(c_k) \approx 0.90$ across seeds. This is the sense in which the nominal slope is a slope on ability whose information is not low, unlike GPCM discrimination whose information carries the squared lever arm $(\theta - \beta)^2$ that is small where the responses concentrate.

C The free-parameter-table invariant and gauge handling

When the item parameters form a free table that can reach the optimal prediction, every parameter gradient is linear in the per-category residual, so at the optimum all residuals and all item gradients vanish together and the recovered item ranks match the true ranks up to the affine gauge, for any readout arrangement; we confirm this numerically for the item table. It does not apply to the encoder's θ , which is not a free table. Because the IRT scale is identified only up to an affine transformation, and any C^1 strictly monotone positive link on discrimination preserves its rank, recovery is measured throughout by the sign-aligned rank correlation rather than by raw magnitude, and only the single global scale is non-identifiable. Magnitude, where reported, is obtained by linking the recovered discrimination to the true scale and reporting the regression slope and RMSE at the optimal scale, which separates a genuine attenuation from the gauge.

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